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Winch Line Pull & Torque Shortfall Worksheet

For maintenance engineers and OEM designers diagnosing DC winches and hoists that cannot lift their rated load

Fill in Steps 1 to 3 at the desk to find the torque the motor must deliver on the worst rope layer. Then carry out Steps 4 and 5 on the winch during a real lift, and use Step 6 to choose the cheapest effective fix.

Step 1 — Define the load and rigging

- [ ] Record the heaviest load to be lifted, m (kg), and its weight W = m × 9.81 (N).
- [ ] Note the number of parts of line: with a snatch block, drum rope tension ≈ W ÷ (1 + sheave efficiency).
- [ ] Count diverting sheaves between drum and load; allow roughly 2–5 % loss per sheave, more for plain-bush sheaves.
- [ ] Confirm the pull is vertical or note the angle; side pull and drag add to the drum load.

Step 2 — Rope layers and effective radius

- [ ] Measure drum barrel radius r0 (mm) and rope diameter d (mm).
- [ ] Count the layers on the drum when the load is at the top of its lift.
- [ ] Effective radius of layer n: rn ≈ r0 + d × (n − 0.5).
- [ ] Drum torque on the top layer: Tdrum = rope tension × rn (convert mm to m).
- [ ] Check winding: loose or crossed turns raise the effective radius and damage the rope.

Step 3 — Drivetrain efficiency and required motor torque

- [ ] Gear efficiency: spur, helical or planetary stages typically 95–98 % each; worm stages roughly 40–90 % depending on ratio and lead angle.
- [ ] A self-locking worm usually has forward efficiency below 50 %, and lower again when starting from rest.
- [ ] Overall efficiency = gear efficiency × drum bearing efficiency × sheave efficiency for each sheave.
- [ ] Required motor torque Tm = Tdrum ÷ (gear ratio × overall efficiency).
- [ ] Compare Tm with the motor's continuous and short-time (S2) torque ratings on the data sheet.

Step 4 — Electrical checks during a real lift

- [ ] Measure battery open-circuit voltage before the test; a 24 V lead-acid pack reads about 25.4 V fully charged.
- [ ] Measure voltage at the motor terminals during the heaviest part of the lift and compare with battery voltage.
- [ ] Measure the voltage drop across each cable run, contactor and bolted joint while current flows; more than about 0.1 V across one joint at around 100 A indicates a poor connection.
- [ ] Log motor current with a DC clamp meter and compare with Tm ÷ Kt plus the no-load current.
- [ ] Check any controller current-limit setting against the motor's permitted peak current.

Step 5 — Brake, mechanical and thermal checks

- [ ] Measure brake coil voltage during the lift and confirm full release; check the air gap against the brake data sheet.
- [ ] After a long pull, compare brake housing and motor frame temperatures; a hot brake means drag.
- [ ] Compare actual on-time per lift with the motor's S2 rating and allow full cool-down between lifts.

- [ ] Remember a hot winding has more resistance (about +0.39 % per °C) and hot ferrite magnets give less torque per amp (about −0.2 % per °C).
- [ ] Check no-load speed against the data sheet: a clearly higher value suggests weakened magnets.

Step 6 — Choose the remedy, cheapest first

- [ ] Restore the supply: charge or replace batteries as a set, remake corroded lugs, upsize cables.
- [ ] Reduce rope layers: shorter rope, longer drum, even winding under tension.
- [ ] Add a snatch block to roughly halve rope tension, accepting half the hook speed.
- [ ] Increase the gear ratio, after confirming gearbox, drum, rope and brake ratings.
- [ ] Replace a self-locking worm with efficient gearing plus a fail-safe brake.
- [ ] Move to a higher system voltage or a larger motor when the load has outgrown the drive.

Example: 500 kg lift, 160 mm drum barrel, 8 mm rope, 60:1 gearing

Layer	Effective radius	Drum torque	Motor torque, 88 % gearing	Motor torque, 45 % worm
1	84 mm	412.0 Nm	7.80 Nm	15.26 Nm
2	92 mm	451.3 Nm	8.55 Nm	16.71 Nm
3	100 mm	490.5 Nm	9.29 Nm	18.17 Nm
4	108 mm	529.7 Nm	10.03 Nm	19.62 Nm
5	116 mm	569.0 Nm	10.78 Nm	21.07 Nm

Values are typical engineering guidance for diagnosis. Never exceed the winch's rated load on any layer, size holding brakes for the top-layer load torque without relying on gear friction, and follow the lifting-equipment rules that apply to your installation. For drive sizing support, contact Globe Scott Motors through www.globescott.net/contact-us/.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.